Python-Plotly

Objective

- Basics of Plotly
- Installation
- Package Structure of Plotly
- Types Of Plots
- Summary

Basics of Plotly

- The Plotly Python library is an interactive open-source library.
- This can be a very helpful tool for data visualization and understanding the data simply and easily.
- plotly graph objects are a high-level interface to plotly which are easy to use.
- It can plot various types of graphs and charts like scatter plots, line charts, bar charts, box plots, histograms, pie charts, etc.

Installation

- use following command to install plotly
  pip install plotly

- To check plotly version
  import plotly
  print(plotly.__version__)

If you are using Anaconda

- Start Navigator.
- In Environments tab checked for installed packages.

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Package Structure of Plotly

OBJECT™ TECHNOLOGIES

There are three main modules in Plotly.

1. plotly.plotly
2. plotly.graph.objects
3. plotly.tools

1. **plotly.plotly** acts as the interface between the local machine and Plotly. It contains functions that require a response from Plotly's server.

2. **plotly.graph_objects** module contains the objects (Figure, layout, data, and the definition of the plots like scatter plot, line chart) that are responsible for creating the plots.
The figures created, manipulated and rendered by the plotly Python library are represented by tree-like data structures which are automatically serialized to JSON for rendering by the Plotly.js JavaScript library.

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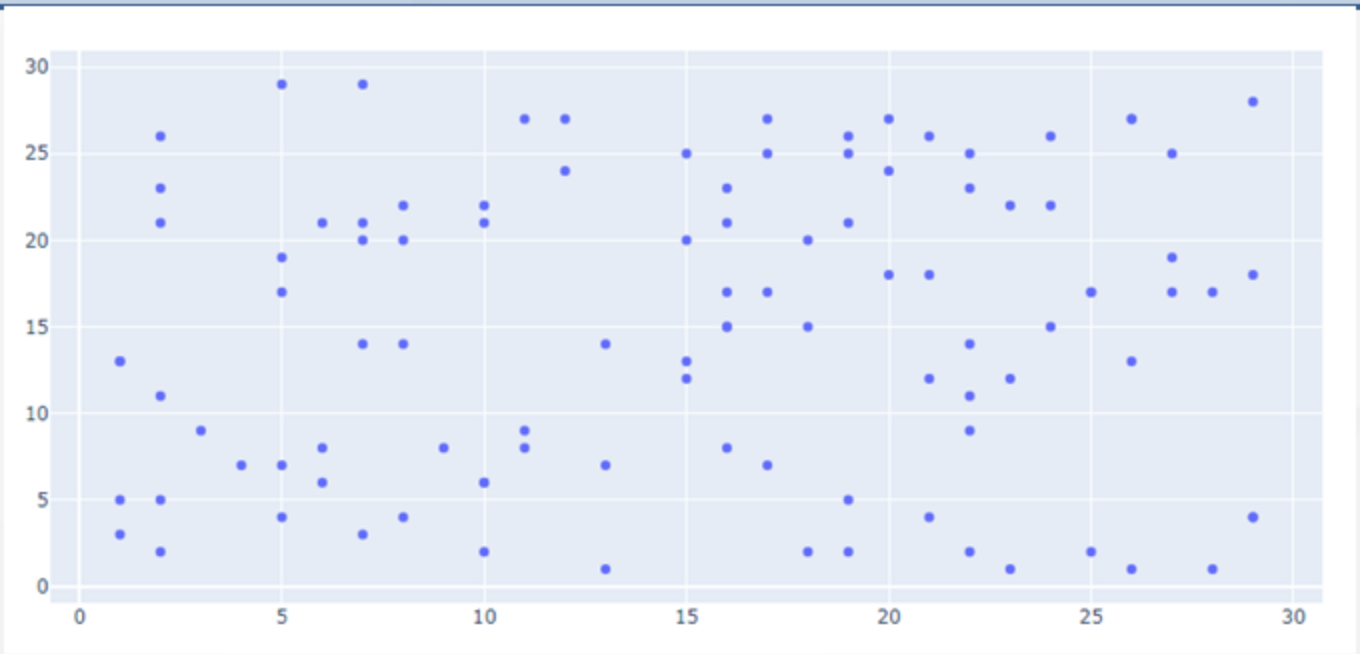
Types Of Plots

- Scatter Chart
- Bar Chart
- Pie Chart

Scatter Chart

```
import plotly
import numpy as np
import plotly.graph_objects as go

xdata= np.random.randint(1,30,100)
ydata= np.random.randint(1,30,100)
fig=go.Figure(data=go.Scatter(x=xdata,y=ydata,mode="markers"))
fig.show()
```



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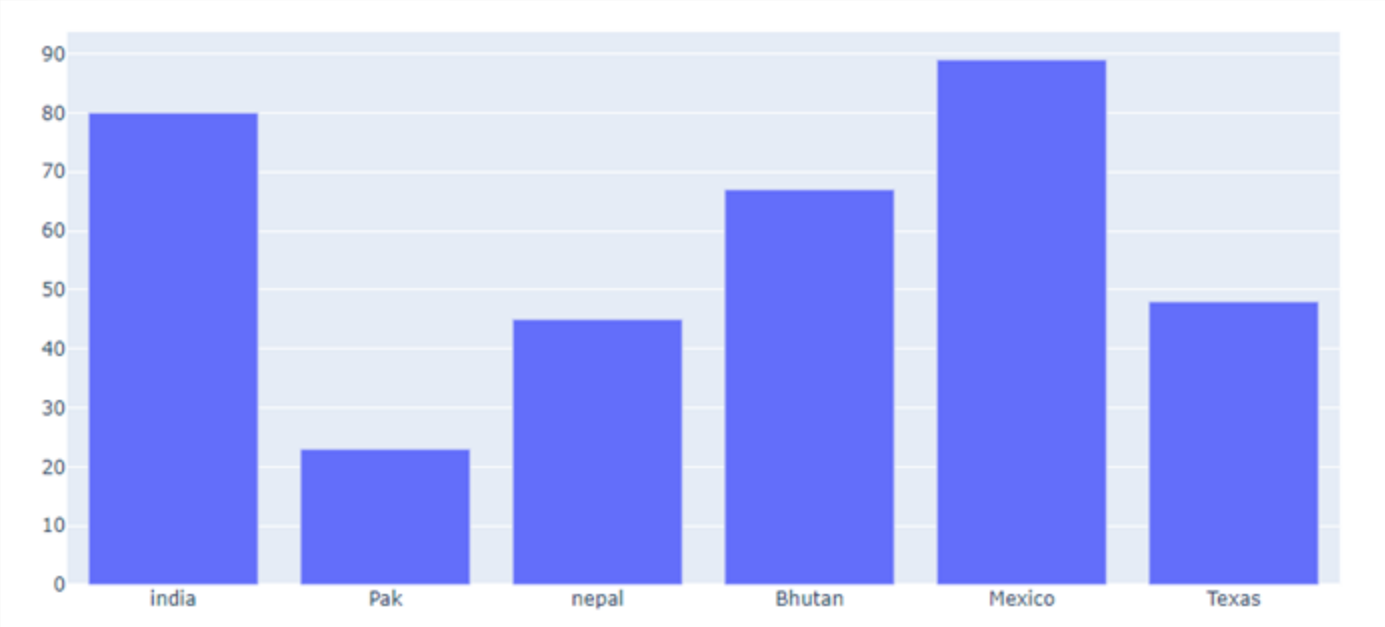
Bar Chart

```
import plotly
import plotly.graph_objects as go

cnt= ["india","Pak","nepal","Bhutan","Mexico","Texas"]
ydata= [80,23,45,67,89,48]
fig=go.Figure([go.Bar(x=cnt,y=ydata)])
fig.show()
```



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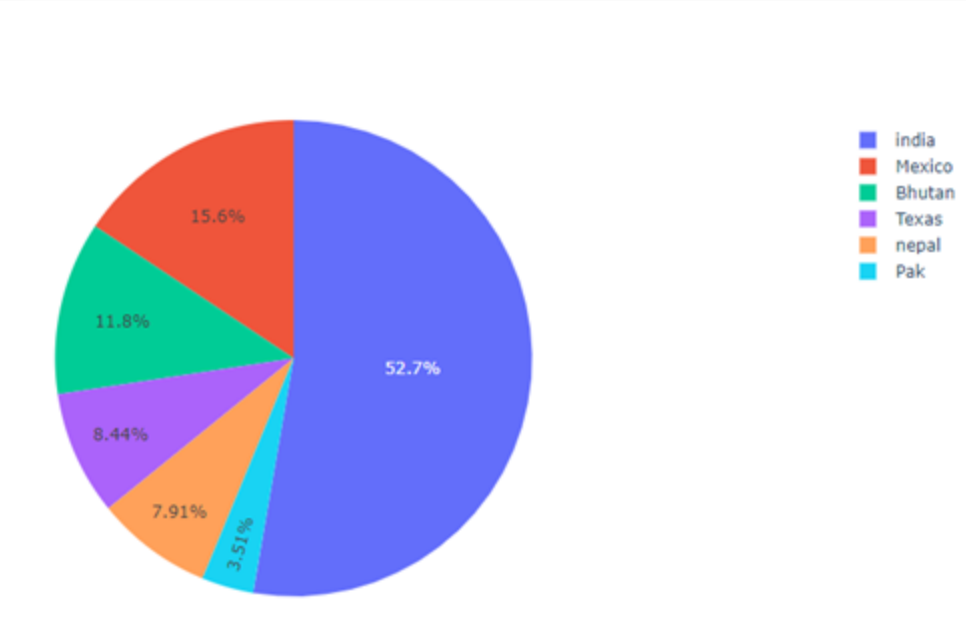
Pie Chart

```
import plotly
import plotly.graph_objects as go

cnt= ["india","Pak","nepal","Bhutan","Mexico","Texas"]
ydata= [3000,200,450,670,890,480]
fig=go.Figure([go.Pie(labels=cnt,values=ydata)])
fig.show()
```



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Thank You